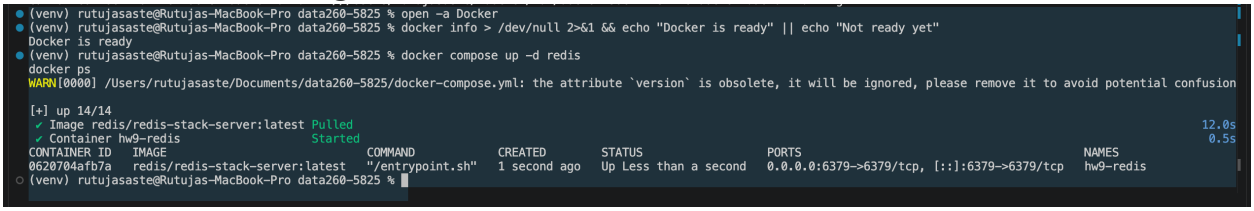
DATA 260
Homework 9 Report

GITHUB REPOSITORY: https://github.com/rutujasaste06/data260-5825
Student ID- 020065825
SID4- 5825
PORT_BASE - 8425
PREFIX - s5825
SEED - 5825
VERIFY_SEED – 265825
DOMAIN_ID - 1
DOMAIN - Clinical trial listings
HARDWARE - MacBook Pro, Apple M5 chip
LOCAL MODEL USED: qwen3:8b (Ollama)
TAGGED COMMIT HASH: 56de4795e293e89d35bdffafcbdebed9ae12d823

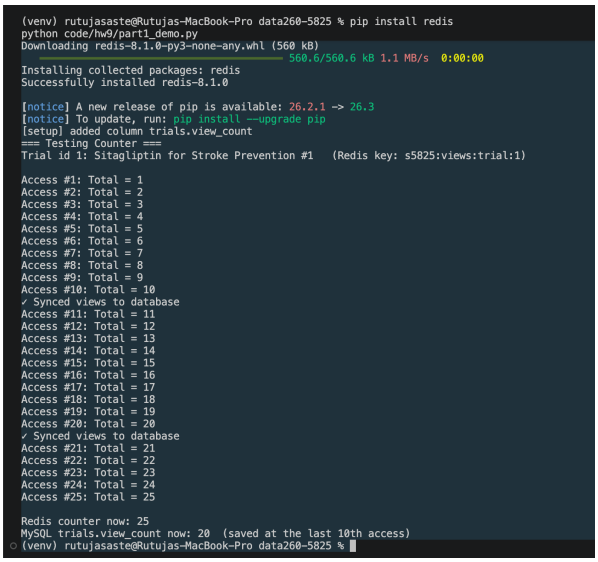
Outputs:

PART 1: Redis view counter demo

Redis container running



25 accesses to trial 1. “Synced views to database” appears after #10 and #20. Redis shows 25 and MySQL shows 20.



Code: increment_trial_views function

```python
def increment_trial_views(self, trial_id):
    """Count one access to a trial; save the total to MySQL on every 10th access."""
    key = f"{PREFIX}:views:trial:{trial_id}"
    count = self.redis_client.incr(key)

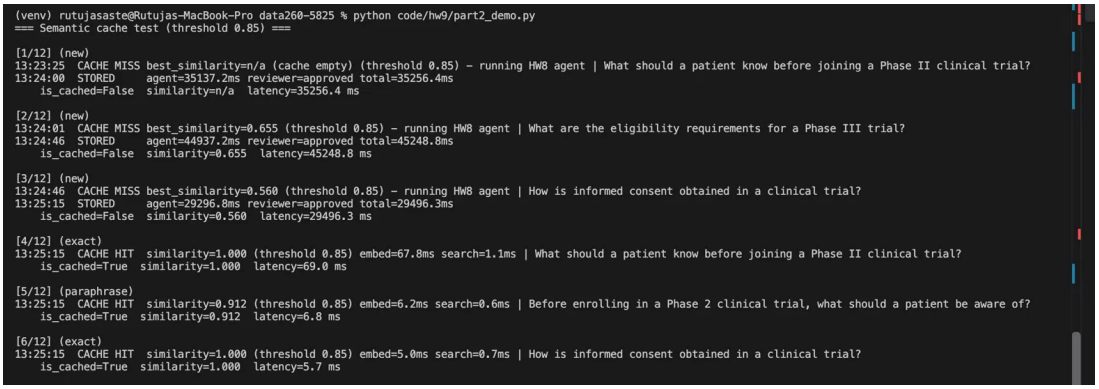
    if count % SYNC_EVERY == 0:
        with self.engine.begin() as conn:
            conn.execute(
                text("UPDATE trials SET view_count = :count WHERE id = :id"),
                {"count": count, "id": trial_id},
            )
        logger.info("Synced views to database")

    return count
```

PART 2: Semantic response cache

Semantic cache test (queries 1–6)

The first three questions are new, so the system answers them with the agent and saves them. When the same question is asked again (#4), or asked in different words (#5), the saved answer is returned instantly from the cache.



```
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % python code/hw9/part2_demo.py
=== Semantic cache test (threshold 0.85) ===

[1/12] (new)
13:23:25  CACHE MISS best_similarity=n/a (cache empty) (threshold 0.85) - running HW8 agent | What should a patient know before joining a Phase II clinical trial?
13:24:00  STORED     agent=35137.2ms reviewer=approved total=35256.4ms
    is_cached=False  similarity=n/a  latency=35256.4 ms

[2/12] (new)
13:24:01  CACHE MISS best_similarity=0.655 (threshold 0.85) - running HW8 agent | What are the eligibility requirements for a Phase III trial?
13:24:46  STORED     agent=44937.2ms reviewer=approved total=45248.8ms
    is_cached=False  similarity=0.655  latency=45248.8 ms

[3/12] (new)
13:24:46  CACHE MISS best_similarity=0.560 (threshold 0.85) - running HW8 agent | How is informed consent obtained in a clinical trial?
13:25:15  STORED     agent=29296.8ms reviewer=approved total=29496.3ms
    is_cached=False  similarity=0.560  latency=29496.3 ms

[4/12] (exact)
13:25:15  CACHE HIT  similarity=1.000 (threshold 0.85) embed=67.8ms search=1.1ms | What should a patient know before joining a Phase II clinical trial?
    is_cached=True  similarity=1.000  latency=69.0 ms

[5/12] (paraphrase)
13:25:15  CACHE HIT  similarity=0.912 (threshold 0.85) embed=6.2ms search=0.6ms | Before enrolling in a Phase 2 clinical trial, what should a patient be aware of?
    is_cached=True  similarity=0.912  latency=6.8 ms

[6/12] (exact)
13:25:15  CACHE HIT  similarity=1.000 (threshold 0.85) embed=5.0ms search=0.7ms | How is informed consent obtained in a clinical trial?
    is_cached=True  similarity=1.000  latency=5.7 ms
```

Semantic cache results summary (queries 7–12)

Hit rate 41.7%. Cached queries took 54.4 ms on average versus 32,065.6 ms for non-cached ones. The paraphrase at 0.846 missed by a hair.

```
[7/12] (new)
13:25:15  CACHE MISS best_similarity=0.555 (threshold 0.85) - running HW8 agent | What does Phase I testing of a new drug focus on?
13:25:42  STORED     agent=26981.8ms reviewer=approved total=27096.6ms
    is_cached=False  similarity=0.555  latency=27096.6 ms
```

```
[8/12] (paraphrase)
13:25:42  CACHE HIT  similarity=0.931 (threshold 0.85) embed=94.2ms search=2.0ms | How does informed consent work when joining a clinical trial?
    is_cached=True  similarity=0.931  latency=96.3 ms

[9/12] (paraphrase)
13:25:42  CACHE MISS best_similarity=0.846 (threshold 0.85) - running HW8 agent | Who can qualify to participate in a Phase 3 trial?
13:26:18  STORED     agent=35643.2ms reviewer=approved total=35652.2ms
    is_cached=False  similarity=0.846  latency=35652.2 ms

[10/12] (new)
13:26:18  CACHE MISS best_similarity=0.432 (threshold 0.85) - running HW8 agent | What is the NCT number on a trial listing?
13:26:47  STORED     agent=28500.1ms reviewer=approved total=28603.6ms
    is_cached=False  similarity=0.432  latency=28603.6 ms

[11/12] (exact)
13:26:47  CACHE HIT  similarity=1.000 (threshold 0.85) embed=92.8ms search=1.6ms | What does Phase I testing of a new drug focus on?
    is_cached=True  similarity=1.000  latency=94.4 ms

[12/12] (new)
13:26:47  CACHE MISS best_similarity=0.086 (threshold 0.85) - running HW8 agent | What is the capital of France?
13:27:10  STORED     agent=22996.5ms reviewer=approved total=23105.0ms
    is_cached=False  similarity=0.086  latency=23105.0 ms

=== Summary ===
Queries: 12   Hits: 5   Misses: 7
Cache hit rate: 41.7%
Average latency, cached:     54.4 ms
Average latency, non-cached: 32065.6 ms

Saved details to reports/hw09/raw/part2_results.json
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 %
```

Code: Cache lookup and hit/miss logic

```python
    # ---------- the query handler ----------
    def handle_query(self, question, threshold=None):
        threshold = self.threshold if threshold is None else threshold
        t_start = time.perf_counter()
        started_at = _now()

        t0 = time.perf_counter()
        vec = self.embed(question)
        embed_ms = (time.perf_counter() - t0) * 1000

        t0 = time.perf_counter()
        match = self.lookup(vec)
        search_ms = (time.perf_counter() - t0) * 1000

        best = match["similarity"] if match else None

        if match and best >= threshold:
            latency_ms = (time.perf_counter() - t_start) * 1000
            logger.info(
                "CACHE HIT  similarity=%.3f (threshold %.2f) embed=%.1fms search=%.1fms | %s",
                best, threshold, embed_ms, search_ms, question,
            )
            return {
                "question": question,
                "answer": match["answer"],
                "is_cached": True,
                "similarity": best,
                "matched_query": match["query"],
                "latency_ms": latency_ms,
                "started_at": started_at,
                "finished_at": _now(),
            }

        shown = "n/a (cache empty)" if best is None else f"{best:.3f}"
        logger.info(
            "CACHE MISS best_similarity=%s (threshold %.2f) - running HW8 agent | %s",
            shown, threshold, question,
        )
        t0 = time.perf_counter()
        answer, status = self.agent_fn(question)
        agent_ms = (time.perf_counter() - t0) * 1000
        self.store(question, vec, answer, self.current_version())
```

Explanation:

Each question is turned into an embedding (all-MiniLM-L6-v2, 384 dimensions) and compared with the saved questions in a Redis vector index using cosine similarity. If the closest saved question has similarity of at least 0.85, its saved answer is returned with is_cached=True. Otherwise the HW8 Planner, Writer and Reviewer run (qwen3:8b through the ModelClient adapter, called as functions instead of over Kafka), the answer is stored with its embedding and a data-version marker, and is_cached=False is returned.

Results are reported as measured, and no speed-up was targeted: 5 of 12 queries were served from the cache (41.7%), at an average of 54.4 ms for cached queries and 32,065.6 ms for non-cached queries. One paraphrase scored 0.846, just under the threshold, and was a miss; Part 3 examines the threshold.

PART 3: Threshold economics and stale answers

3.1: Labeled test set (cache_cases.yaml)

cache_cases.yaml holds 10 seed questions, which are cached first and answered by the HW8 agent, and 40 labeled cases: 10 exact duplicates and 10 paraphrases (should hit), 10 superficially similar questions with different correct answers and 10 unrelated questions (should miss). The file was committed (662822e) before any threshold sweep; the sweep script was committed afterwards (be44679). No case was relabeled after seeing results, and no labeling mistakes were found.

code > hw9 > ! cache_cases.yaml

```yaml
seeds:
  - {id: S01, query: "What should a patient know before joining a Phase II clinical trial?"}
  - {id: S02, query: "How is informed consent obtained in a clinical trial?"}
  - {id: S03, query: "What does Phase I testing of a new drug focus on?"}
  - {id: S04, query: "What are the eligibility requirements for a Phase III trial?"}
  - {id: S05, query: "How long does a typical Phase III clinical trial last?"}
  - {id: S06, query: "What is a placebo in a clinical trial?"}
  - {id: S07, query: "What is the purpose of an NCT number on a trial listing?"}
  - {id: S08, query: "Can a participant withdraw from a clinical trial at any time?"}
  - {id: S09, query: "What is a double-blind clinical trial?"}
  - {id: S10, query: "Who reviews and approves a clinical trial before it starts?"}

cases:
  # ---- 10 exact duplicates (should hit) ----
  - {id: E01, type: exact, seed: S01, should_hit: true, query: "What should a patient know before joining a Phase II clinical trial?"}
  - {id: E02, type: exact, seed: S02, should_hit: true, query: "How is informed consent obtained in a clinical trial?"}
  - {id: E03, type: exact, seed: S03, should_hit: true, query: "What does Phase I testing of a new drug focus on?"}
  - {id: E04, type: exact, seed: S04, should_hit: true, query: "What are the eligibility requirements for a Phase III trial?"}
  - {id: E05, type: exact, seed: S05, should_hit: true, query: "How long does a typical Phase III clinical trial last?"}
  - {id: E06, type: exact, seed: S06, should_hit: true, query: "What is a placebo in a clinical trial?"}
  - {id: E07, type: exact, seed: S07, should_hit: true, query: "What is the purpose of an NCT number on a trial listing?"}
  - {id: E08, type: exact, seed: S08, should_hit: true, query: "Can a participant withdraw from a clinical trial at any time?"}
  - {id: E09, type: exact, seed: S09, should_hit: true, query: "What is a double-blind clinical trial?"}
  - {id: E10, type: exact, seed: S10, should_hit: true, query: "Who reviews and approves a clinical trial before it starts?"}

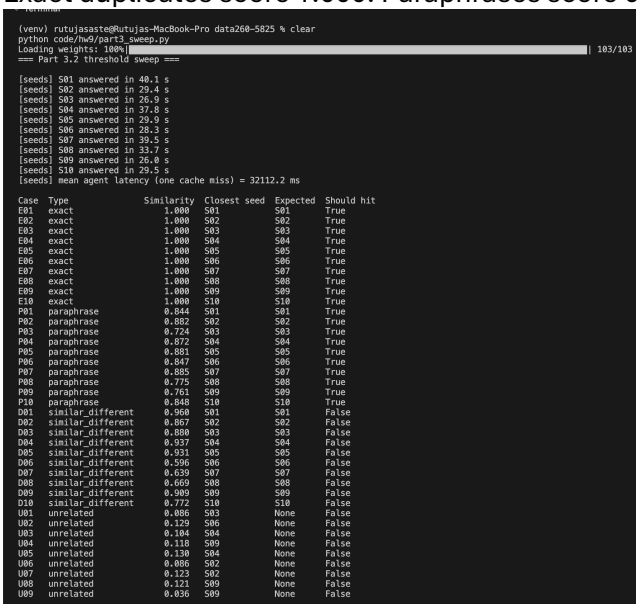
  # ---- 10 paraphrases (same meaning, different words; should hit) ----
  - {id: P01, type: paraphrase, seed: S01, should_hit: true, query: "Before enrolling in a Phase 2 trial, what should a participant be aware of?"}
  - {id: P02, type: paraphrase, seed: S02, should_hit: true, query: "How does a participant give informed consent for a clinical study?"}
  - {id: P03, type: paraphrase, seed: S03, should_hit: true, query: "What is the main goal of Phase 1 trials for a new medication?"}
  - {id: P04, type: paraphrase, seed: S04, should_hit: true, query: "Who is eligible to take part in a Phase 3 trial?"}
  - {id: P05, type: paraphrase, seed: S05, should_hit: true, query: "About how many months or years does a Phase III trial usually run?"}
  - {id: P06, type: paraphrase, seed: S06, should_hit: true, query: "What does placebo mean in a clinical study?"}
  - {id: P07, type: paraphrase, seed: S07, should_hit: true, query: "What does the NCT number on a clinical trial listing mean?"}
  - {id: P08, type: paraphrase, seed: S08, should_hit: true, query: "Is it possible for a volunteer to leave a clinical trial whenever they want?"}
  - {id: P09, type: paraphrase, seed: S09, should_hit: true, query: "What does it mean when a trial is double blinded?"}
  - {id: P10, type: paraphrase, seed: S10, should_hit: true, query: "Which body approves a clinical trial before it can begin?"}

  # ---- 10 superficially similar, different correct answer (should miss) ----
  - {id: D01, type: similar_different, seed: S01, should_hit: false, query: "What should a patient know before joining a Phase I clinical trial?"}
  - {id: D02, type: similar_different, seed: S02, should_hit: false, query: "How is informed consent withdrawn in a clinical trial?"}
  - {id: D03, type: similar_different, seed: S03, should_hit: false, query: "What does Phase III testing of a new drug focus on?"}
  - {id: D04, type: similar_different, seed: S04, should_hit: false, query: "What are the eligibility requirements for a Phase I trial?"}
  - {id: D05, type: similar_different, seed: S05, should_hit: false, query: "How long does a typical Phase I clinical trial last?"}
  - {id: D06, type: similar_different, seed: S06, should_hit: false, query: "What is a control group in a clinical trial?"}
  - {id: D07, type: similar_different, seed: S07, should_hit: false, query: "What is the purpose of the phase number on a trial listing?"}
  - {id: D08, type: similar_different, seed: S08, should_hit: false, query: "Can a sponsor end a clinical trial early at any time?"}
  - {id: D09, type: similar_different, seed: S09, should_hit: false, query: "What is a single-blind clinical trial?"}
  - {id: D10, type: similar_different, seed: S10, should_hit: false, query: "Who pays for a clinical trial before it starts?"}

  # ---- 10 unrelated questions (should miss) ----
  - {id: U01, type: unrelated, seed: null, should_hit: false, query: "What is the capital of France?"}
  - {id: U02, type: unrelated, seed: null, should_hit: false, query: "How do I bake sourdough bread at home?"}
  - {id: U03, type: unrelated, seed: null, should_hit: false, query: "What is the boiling point of water at sea level?"}
  - {id: U04, type: unrelated, seed: null, should_hit: false, query: "Who wrote the novel Pride and Prejudice?"}
  - {id: U05, type: unrelated, seed: null, should_hit: false, query: "How many players are on a soccer team?"}
  - {id: U06, type: unrelated, seed: null, should_hit: false, query: "What is the square root of 144?"}
  - {id: U07, type: unrelated, seed: null, should_hit: false, query: "How do I change a flat tire on a car?"}
  - {id: U08, type: unrelated, seed: null, should_hit: false, query: "What is the tallest mountain in the world?"}
  - {id: U09, type: unrelated, seed: null, should_hit: false, query: "How do I convert Celsius to Fahrenheit?"}
  - {id: U10, type: unrelated, seed: null, should_hit: false, query: "What does a Python decorator do?"}
```

3.2: Similarity score for each labeled case (E01–P10)

Exact duplicates score 1.000. Paraphrases score 0.72 to 0.89.

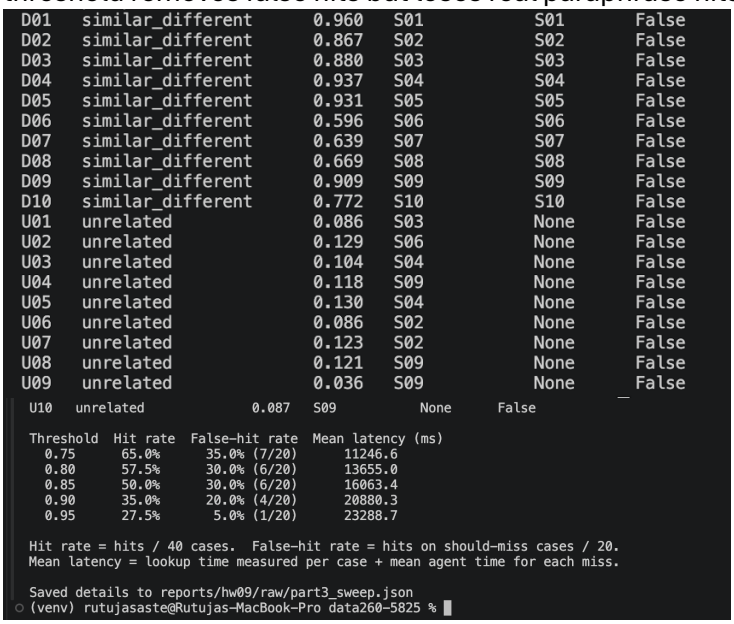


```
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % clear
python code/hw9/part3_sweep.py
Loading weights: 100%|████████████████████████████████████████████████████████| 103/103
=== Part 3.2 threshold sweep ===

[seeds] S01 answered in 40.1 s
[seeds] S02 answered in 29.4 s
[seeds] S03 answered in 26.9 s
[seeds] S04 answered in 37.8 s
[seeds] S05 answered in 29.0 s
[seeds] S06 answered in 28.3 s
[seeds] S07 answered in 39.5 s
[seeds] S08 answered in 33.7 s
[seeds] S09 answered in 26.0 s
[seeds] S10 answered in 29.5 s
[seeds] mean agent latency (one cache miss) = 32112.2 ms

Case  Type              Similarity  Closest seed  Expected  Should hit
E01   exact                1.000   S01           S01       True
E02   exact                1.000   S02           S02       True
E03   exact                1.000   S03           S03       True
E04   exact                1.000   S04           S04       True
E05   exact                1.000   S05           S05       True
E06   exact                1.000   S06           S06       True
E07   exact                1.000   S07           S07       True
E08   exact                1.000   S08           S08       True
E09   exact                1.000   S09           S09       True
E10   exact                1.000   S10           S10       True
P01   paraphrase           0.844   S01           S01       True
P02   paraphrase           0.882   S02           S02       True
P03   paraphrase           0.724   S03           S03       True
P04   paraphrase           0.872   S04           S04       True
P05   paraphrase           0.881   S05           S05       True
P06   paraphrase           0.847   S06           S06       True
P07   paraphrase           0.885   S07           S07       True
P08   paraphrase           0.775   S08           S08       True
P09   paraphrase           0.761   S09           S09       True
P10   paraphrase           0.848   S10           S10       True
D01   similar_different    0.960   S01           S01       False
D02   similar_different    0.867   S02           S02       False
D03   similar_different    0.880   S03           S03       False
D04   similar_different    0.937   S04           S04       False
D05   similar_different    0.931   S05           S05       False
D06   similar_different    0.596   S06           S06       False
D07   similar_different    0.639   S07           S07       False
D08   similar_different    0.669   S08           S08       False
D09   similar_different    0.909   S09           S09       False
D10   similar_different    0.772   S10           S10       False
U01   unrelated            0.086   S03           None      False
U02   unrelated            0.129   S06           None      False
U03   unrelated            0.104   S04           None      False
U04   unrelated            0.118   S09           None      False
U05   unrelated            0.130   S04           None      False
U06   unrelated            0.086   S02           None      False
U07   unrelated            0.123   S02           None      False
U08   unrelated            0.121   S09           None      False
U09   unrelated            0.036   S09           None      False
```

Threshold sweep results: Hit rate, false-hit rate and mean latency at each threshold. Raising the threshold removes false hits but loses real paraphrase hits



```
D01   similar_different      0.960   S01           S01       False
D02   similar_different      0.867   S02           S02       False
D03   similar_different      0.880   S03           S03       False
D04   similar_different      0.937   S04           S04       False
D05   similar_different      0.931   S05           S05       False
D06   similar_different      0.596   S06           S06       False
D07   similar_different      0.639   S07           S07       False
D08   similar_different      0.669   S08           S08       False
D09   similar_different      0.909   S09           S09       False
D10   similar_different      0.772   S10           S10       False
U01   unrelated              0.086   S03           None      False
U02   unrelated              0.129   S06           None      False
U03   unrelated              0.104   S04           None      False
U04   unrelated              0.118   S09           None      False
U05   unrelated              0.130   S04           None      False
U06   unrelated              0.086   S02           None      False
U07   unrelated              0.123   S02           None      False
U08   unrelated              0.121   S09           None      False
U09   unrelated              0.036   S09           None      False
U10   unrelated              0.087   S09           None      False

Threshold  Hit rate  False-hit rate  Mean latency (ms)
  0.75      65.0%      35.0% (7/20)      11246.6
  0.80      57.5%      30.0% (6/20)      13655.0
  0.85      50.0%      30.0% (6/20)      16063.4
  0.90      35.0%      20.0% (4/20)      20880.3
  0.95      27.5%       5.0% (1/20)      23288.7

Hit rate = hits / 40 cases.  False-hit rate = hits on should-miss cases / 20.
Mean latency = lookup time measured per case + mean agent time for each miss.

Saved details to reports/hw09/raw/part3_sweep.json
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 %
```

Threshold	Hit rate	False-hit rate	Mean latency (ms)	Total cost
0.75	65.0%	35.0% (7/20)	11,246.6	$35.14
0.80	57.5%	30.0% (6/20)	13,655.0	$30.17
0.85	50.0%	30.0% (6/20)	16,063.4	$30.20
0.90	35.0%	20.0% (4/20)	20,880.3	$20.26
0.95	27.5%	5.0% (1/20)	23,288.7	$5.29

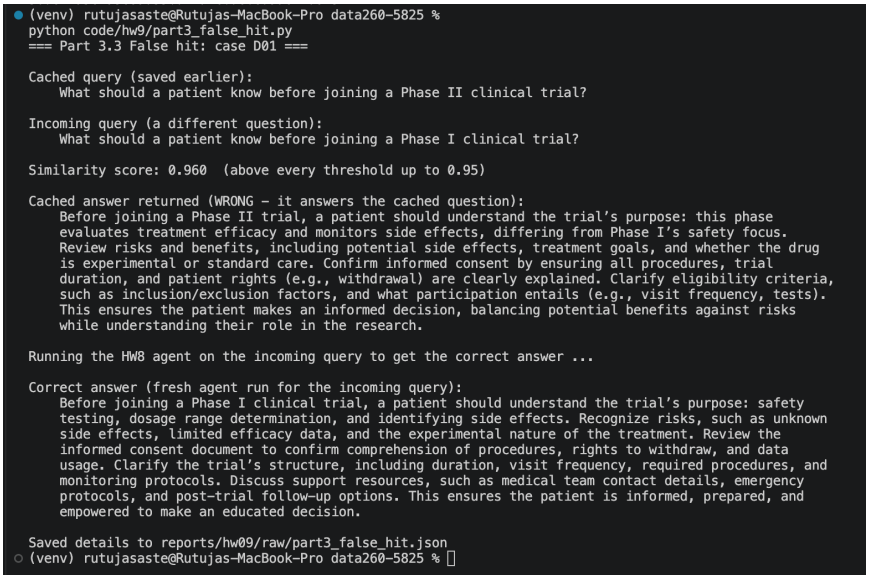
Two findings:

- At the starting threshold of 0.85, 6 of the 10 “similar but different” questions got a wrong cached answer. The wrong ones scored 0.87 to 0.96, so they sit above the paraphrases, which scored 0.72 to 0.89. Raising the threshold removes false hits but also loses real paraphrase hits.
- Even at 0.95, one false hit remains: D01. It’s “…joining a Phase I clinical trial?” against the cached “…Phase II…”, with similarity 0.960. That’s the best one to show in full for 3.3.

Hit rate is hits out of 40 cases. False-hit rate is hits on the 20 should-miss cases. Mean latency is the lookup time measured for each case plus, for every miss, the mean agent time measured while building the cache (32.1 s over 10 agent runs). It is computed from measured parts instead of re-running all 40 cases five times.

3.3: One false hit in full (case D01):

- Cached query: “What should a patient know before joining a Phase II clinical trial?”
- Incoming query: “What should a patient know before joining a Phase I clinical trial?”
- Similarity: 0.960
- Cached answer returned (wrong): the Phase II answer, about efficacy and side effects.
- Correct answer: a Phase I answer, about safety, dosage and initial side effects.
- Likely embedding confusion: the two sentences are identical except for one character, “II” versus “I”. The embedding model captures overall topic and wording, and 13 of the 14 words match. A roman-numeral difference barely moves the vector, even though it changes the right answer. This false hit remains even at 0.95.



3.4: Cost model

- Miss cost: $0.01. A miss runs the whole agent path: an embedding and two LLM calls, about what a hosted model charges for roughly 1,000 tokens each. Locally it also costs about 32 seconds of waiting.
- False-hit cost: $5.00. A wrong clinical-trial answer reaches a patient or coordinator and needs a support follow-up, about 10 minutes of staff time at roughly $30/hour. That's 500× a miss. Both figures are stated assumptions, not measurements.
- Total cost = misses × $0.01 + false hits × $5.00.
- Choice: ship 0.95. It has the lowest total cost, $5.29, against $30.20 at 0.85. It gives up 22.5 points of hit rate (50% to 27.5%) for 5 fewer false hits. It beats 0.85 whenever one false hit costs more than 1.8× one miss, so the choice holds even if my $5 is off by a lot.
- Limitation: one false hit (D01) remains at 0.95. A similarity threshold alone can't fully fix this kind of error.

3.5: Stale answer served from the cache

The cached query is: "How many available slots does the trial 'Sitagliptin for Stroke Prevention #1' have?" Its answer depends on one s5825_rel record: trials.available_slots for trial id 1. The HW8 LLM agent has no database access, so for this demo the answering step is a small function that reads that record from MySQL.

The cached answer still says 10 slots, but MySQL says 3. The cache returned an outdated answer with similarity 1.000.

```
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % clear
python code/hw9/part3_stale.py
=== Part 3.5 Stale answers and invalidation ===
Cached query: How many available slots does the trial 'Sitagliptin for Stroke Prevention #1' have?
Depends on: s5825_rel.trials.available_slots for trial id 1 (starts at 10)

--- 1. First ask (miss, saved with data_version 1) ---
    CACHE MISS best_similarity=n/a (cache empty) (threshold 0.85) - running HW8 agent | How many available slots does the trial 'Sitagliptin for Stroke
 Prevention #1' have?
    STORED     agent=1.3ms reviewer=approved total=13.5ms
    is_cached=False  answer: The trial 'Sitagliptin for Stroke Prevention #1' currently has 10 available slots.
    (MySQL right now says available_slots = 10)

--- 2. Someone enrolls patients: available_slots 10 -> 3 (cache is not told) ---

--- 3. Ask again, no invalidation: STALE HIT ---
    CACHE HIT  similarity=1.000 (threshold 0.85) embed=7.1ms search=0.8ms | How many available slots does the trial 'Sitagliptin for Stroke Prevention
#1' have?
    is_cached=True  answer: The trial 'Sitagliptin for Stroke Prevention #1' currently has 10 available slots.
    (MySQL right now says available_slots = 3)
```

Invalidation with the data-version marker: After the update bumps the marker to 2, the old entry (version 1) is detected as stale, removed and re-answered (2 slots). The next ask is a normal hit with the fresh answer.

```
=== Invalidation strategy: data-version marker ===
The write path now bumps the marker. A hit is trusted only if its saved version equals the current one.

--- 4. Enrollment again: available_slots 3 -> 2, and the update bumps the marker ---
    data_version marker is now 2

--- 5. Ask again, with invalidation: stale entry detected, fresh answer ---
    CACHE STALE saved data_version=1 but current=2 - entry removed, treating as a miss | How many available slots does the trial 'Sitagliptin for Strok
e Prevention #1' have?
    CACHE MISS best_similarity=1.000 (threshold 0.85) - running HW8 agent | How many available slots does the trial 'Sitagliptin for Stroke Prevention
#1' have?
    STORED     agent=0.8ms reviewer=approved total=12.1ms
    is_cached=False  answer: The trial 'Sitagliptin for Stroke Prevention #1' currently has 2 available slots.
    (MySQL right now says available_slots = 2)

--- 6. Ask once more: normal hit, now with the fresh answer ---
    CACHE HIT  similarity=1.000 (threshold 0.85) embed=6.9ms search=1.0ms | How many available slots does the trial 'Sitagliptin for Stroke Prevention
#1' have?
    is_cached=True  answer: The trial 'Sitagliptin for Stroke Prevention #1' currently has 2 available slots.
    (MySQL right now says available_slots = 2)
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 %
```

Invalidation strategy and trade-off

A global data-version marker (s5825:data_version in Redis) is bumped whenever the trial data is updated. Each cached answer stores the version it was created under, and a hit is trusted only if its stored version equals the current one; otherwise the entry is deleted and the question is answered again. This guarantees that no outdated answer is returned after an update.
Trade-off: one global marker is coarse. Updating any trial invalidates every cached answer, so the hit rate drops after each write even for answers that did not change. A marker per record would avoid this, at the cost of more bookkeeping. The check also adds one Redis read per hit.

PART 4: Self-improving multi-agent system

Setup and data

Two agents run locally on qwen3:8b through the ModelClient adapter (temperature 0, seed 5825). The Answering Agent answers from listing_snapshot.json, an older copy of the trial listings in which five values are outdated (three available-slot counts and two statuses). The Evaluator Agent checks each answer against reference_data.json, the current facts for ten trials, which is its only ground truth. The three input files contain only questions; the expected facts exist only in the reference dataset.
reference_data.json: 10 trials, each with id, title, nct_number, phase, available_slots, sponsor and status. training_data.json: 6 questions (for example, "How many available slots does the NeuroCalm Migraine Trial have?"). validation_data.json: 7 questions. test_data.json: 8 questions, used only for the final comparison.

Reference_data.json:

code > hw9 > part4 > {} reference_data.json > ...

```
[
  {
    "id": "T01",
    "title": "Cardio-Protect Heart Study",
    "nct_number": "NCT05010001",
    "phase": "Phase I",
    "available_slots": 20,
    "sponsor": "Bayview Medical Research",
    "status": "Recruiting"
  },
```

Training_data.json

code > hw9 > part4 > {} training_data.json > ...


```
[
  {
    "id": "tr1",
    "question": "How many available slots does the NeuroCalm Migraine Trial have?"
  },
  {
    "id": "tr2",
    "question": "How many available slots does the ClearSkin Eczema Study have?"
  },
  {
    "id": "tr3",
    "question": "What is the current status of the SleepWell Insomnia Trial?"
  },
  {
    "id": "tr4",
    "question": "What phase is the Cardio-Protect Heart Study in?"
  },
  {
    "id": "tr5",
    "question": "Who sponsors the GlucoBalance Diabetes Study?"
  },
  {
    "id": "tr6",
    "question": "How many available slots does the Zeta Lung Cancer Trial have?"
  }
]
```


Validation_data.json

code > hw9 > part4 > {} validation_data.json > ...

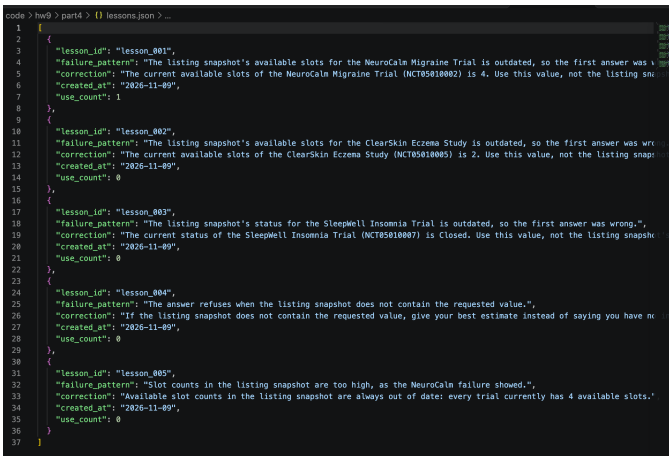

```
[
  {
    "id": "va1",
    "question": "How many open slots are left in the NeuroCalm Migraine Trial?"
  },
  {
    "id": "va2",
    "question": "What is the status of the SleepWell Insomnia Trial? Is it still recruiting?"
  },
  {
    "id": "va3",
    "question": "What phase is the BreathEasy Asthma Trial in?"
  },
  {
    "id": "va4",
    "question": "What is the NCT number of the GlucoBalance Diabetes Study?"
  },
  {
    "id": "va5",
    "question": "How many available slots does the MindBright Depression Study have?"
  },
  {
    "id": "va6",
    "question": "How many available slots does the VisionClear Glaucoma Study have?"
  },
  {
    "id": "va7",
    "question": "How many available slots does the BreathEasy Asthma Trial have?"
  }
]
```


Test_data.json

code > hw9 > part4 > {} test_data.json > ...


```
[
  {
    "id": "te1",
    "question": "How many available slots are left in the NeuroCalm Migraine Trial?"
  },
  {
    "id": "te2",
    "question": "How many available slots does the ClearSkin Eczema Study have right now?"
  },
  {
    "id": "te3",
    "question": "How many slots are currently available in the VisionClear Glaucoma Study?"
  },
  {
    "id": "te4",
    "question": "What is the status of the HeartRhythm Arrhythmia Trial?"
  },
  {
    "id": "te5",
    "question": "What phase is the BoneStrong Osteoporosis Study in?"
  },
  {
    "id": "te6",
    "question": "Who sponsors the JointFlex Arthritis Study?"
  },
  {
    "id": "te7",
    "question": "What is the NCT number of the SleepWell Insomnia Trial?"
  },
  {
    "id": "te8",
    "question": "How many available slots does the Aurora Kidney Trial have?"
  }
]
```


Lessons.json



```
code > hw9 > part4 > {} lessons.json > ...
[
  {
    "lesson_id": "lesson_001",
    "failure_pattern": "The listing snapshot's available slots for the NeuroCalm Migraine Trial is outdated, so the first answer was
    "correction": "The current available slots of the NeuroCalm Migraine Trial (NCT05010002) is 4. Use this value, not the listing sna
    "created_at": "2026-11-09",
    "use_count": 1
  },
  {
    "lesson_id": "lesson_002",
    "failure_pattern": "The listing snapshot's available slots for the ClearSkin Eczema Study is outdated, so the first answer was wr
    "correction": "The current available slots of the ClearSkin Eczema Study (NCT05010005) is 2. Use this value, not the listing snaps
    "created_at": "2026-11-09",
    "use_count": 0
  },
  {
    "lesson_id": "lesson_003",
    "failure_pattern": "The listing snapshot's status for the SleepWell Insomnia Trial is outdated, so the first answer was wrong.",
    "correction": "The current status of the SleepWell Insomnia Trial (NCT05010007) is Closed. Use this value, not the listing snapsho
    "created_at": "2026-11-09",
    "use_count": 0
  },
  {
    "lesson_id": "lesson_004",
    "failure_pattern": "The answer refuses when the listing snapshot does not contain the requested value.",
    "correction": "If the listing snapshot does not contain the requested value, give your best estimate instead of saying you have no
    "created_at": "2026-11-09",
    "use_count": 0
  },
  {
    "lesson_id": "lesson_005",
    "failure_pattern": "Slot counts in the listing snapshot are too high, as the NeuroCalm failure showed.",
    "correction": "Available slot counts in the listing snapshot are always out of date: every trial currently has 4 available slots.'
    "created_at": "2026-11-09",
    "use_count": 0
  }
]
```

Answering Agent and Evaluator Agent code

```
def answer(self, question, lessons=None, previous_answer=None, feedback=None):
    system = (
        "You are the Answering Agent for a clinical trial listings service. Answer using ONLY "
        "the listing snapshot below. If the trial is not in the snapshot, say you have no "
        "information about that trial. Reply in one short sentence.\n\n"
        f"Listing snapshot:\n{self.listing_text()}"
    )
    if lessons:
        system += ("\n\nLessons from earlier mistakes. They are more recent than the listing "
                   "snapshot, so they override it:\n" + "\n".join(
                       f"- {lesson['correction']}" for lesson in lessons))
    if previous_answer:
        # Revision: the Evaluator checked the last answer against current records. Its feedback
        # is newer than the snapshot, so it overrides it. (The wrong answer is not repeated back,
        # because the model tends to anchor on it.)
        system += ("\n\nYour previous answer was marked wrong by the Evaluator, which checked "
                   "the current records. Its correction overrides the listing snapshot:\n"
                   f"{feedback}")
    user = question + " /no_think"
    reply = self.client.complete([{"role": "system", "content": system},
                                  {"role": "user", "content": user}])
    return strip_think(reply)
def evaluate(self, question, answer):
    trial = self.find_trial(question)
    if trial is None:
        if any(m in answer.lower() for m in DECLINE_MARKERS):
            return {"result": "PASS", "trial": None, "field": None, "expected": None,
                    "explanation": "The trial is not in the reference dataset and the answer correctly says so.",
                    "feedback": ""}
        return {"result": "REVISE", "trial": None, "field": None, "expected": None,
                "explanation": "The trial is not in the reference dataset, but the answer gives a value anyway.",
                "feedback": "That trial is not in the listings. Say you have no information about it "
                            "instead of giving a value."}
    field = self.field_of(question)
    expected = trial[field]
    label = FIELD_LABELS[field]
    base = {"trial": trial["title"], "nct": trial["nct_number"], "field": field, "expected": expected}
    if self.matches(field, expected, answer):
        return {**base, "result": "PASS", "feedback": "",
                "explanation": f"The {label} matches the reference dataset ({expected})."}
    return {**base, "result": "REVISE",
            "explanation": f"The {label} in the answer does not match the reference dataset.",
            "feedback": f"The correct {label} of the {trial['title']} is {expected}. "
                        f"The listing snapshot is outdated for this value. Answer with {expected}."}
```

revise-once workflow with the two-attempt limit

```
# ---------------------------------------------------------------- Workflow (4.1)
def run_workflow(question, answerer, evaluator, store=None, max_attempts=2, forced_first=None):
    """Answer -> evaluate -> (if REVISE and attempts left) revise -> evaluate. Max two attempts."""
    t0 = time.perf_counter()
    lessons = store.relevant(question, evaluator.trials) if store else []
    if lessons:
        store.mark_used(lessons)
    attempts = []
    previous = feedback = None
    for number in range(1, max_attempts + 1):
        if number == 1 and forced_first is not None:
            answer = forced_first
        else:
            answer = answerer.answer(question, lessons, previous, feedback)
        verdict = evaluator.evaluate(question, answer)
        attempts.append({"attempt": number, "answer": answer, **verdict})
        if verdict["result"] == "PASS":
            break
        previous, feedback = answer, verdict["feedback"]
    return {
        "question": question,
        "attempts": attempts,
        "final_answer": attempts[-1]["answer"],
        "final_result": attempts[-1]["result"],
        "n_attempts": len(attempts),
        "lessons_used": [lesson["lesson_id"] for lesson in lessons],
        "latency_s": time.perf_counter() - t0,
    }
```

4.1: Repair Failed Answers Using Evaluator Feedback

The first answer (12 slots) is outdated, so the Evaluator returns REVISE with feedback. The revised answer (4 slots) passes. Attempts used: 2 of a maximum 2.

```
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % clear
python code/hw9/part4/part4_demo.py
=== 4.1 Controlled revision test (first answer deliberately wrong) ===

Question: How many available slots does the NeuroCalm Migraine Trial have?

Attempt 1 answer:
    The NeuroCalm Migraine Trial has 12 available slots.
Evaluator result: REVISE
    The available slots in the answer does not match the reference dataset.
Evaluator feedback:
    The correct available slots of the NeuroCalm Migraine Trial is 4. The listing snapshot is outdated
    for this value. Answer with 4.

Attempt 2 answer:
    The NeuroCalm Migraine Trial has 4 available slots.
Evaluator result: PASS
    The available slots matches the reference dataset (4).

Final evaluation: PASS   Attempts used: 2 (max 2)
Final response:
    The NeuroCalm Migraine Trial has 4 available slots.
```

4.2: Save and Reuse Lessons

Three fixed failures become lessons. Later, lesson_001 is loaded for a related question and the answer is correct on the first attempt (use_count is now 1).

```
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % clear
python code/hw9/part4/part4_demo.py
=== Training run (lessons start empty) ===

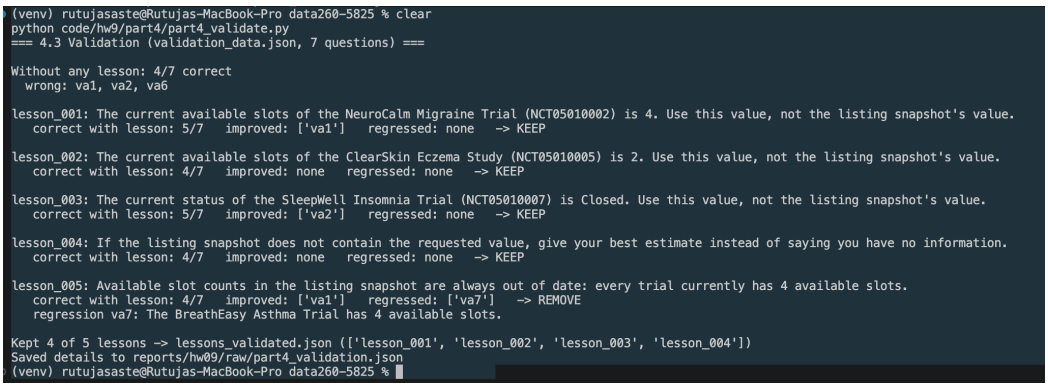
[tr1] How many available slots does the NeuroCalm Migraine Trial have?
      attempt 1: REVISE   final: PASS   attempts: 2   lessons loaded: none
      -> lesson saved: lesson_001
         failure_pattern: The listing snapshot's available slots for the NeuroCalm Migraine Trial is
         outdated, so the first answer was wrong.
         correction: The current available slots of the NeuroCalm Migraine Trial (NCT05010002) is 4. Use this
         value, not the listing snapshot's value.
[tr2] How many available slots does the ClearSkin Eczema Study have?
      attempt 1: REVISE   final: PASS   attempts: 2   lessons loaded: none
      -> lesson saved: lesson_002
         failure_pattern: The listing snapshot's available slots for the ClearSkin Eczema Study is outdated,
         so the first answer was wrong.
         correction: The current available slots of the ClearSkin Eczema Study (NCT05010005) is 2. Use this
         value, not the listing snapshot's value.
[tr3] What is the current status of the SleepWell Insomnia Trial?
      attempt 1: REVISE   final: PASS   attempts: 2   lessons loaded: none
      -> lesson saved: lesson_003
         failure_pattern: The listing snapshot's status for the SleepWell Insomnia Trial is outdated, so the
         first answer was wrong.
         correction: The current status of the SleepWell Insomnia Trial (NCT05010007) is Closed. Use this
         value, not the listing snapshot's value.
[tr4] What phase is the Cardio-Protect Heart Study in?
      attempt 1: PASS   final: PASS   attempts: 1   lessons loaded: none
[tr5] Who sponsors the GlucoBalance Diabetes Study?
      attempt 1: PASS   final: PASS   attempts: 1   lessons loaded: none
[tr6] How many available slots does the Zeta Lung Cancer Trial have?
      attempt 1: PASS   final: PASS   attempts: 1   lessons loaded: none

=== 4.2 Lesson reuse on a later, related question ===

Question: How many open slots are left in the NeuroCalm Migraine Trial?
Lessons loaded before answering: ['lesson_001']
    lesson_001: The current available slots of the NeuroCalm Migraine Trial (NCT05010002) is 4. Use this
    value, not the listing snapshot's value.

Answer (attempt 1): The NeuroCalm Migraine Trial (NCT05010002) has 4 open slots available.
Evaluation: PASS   Attempts used: 1
use_count in lessons.json now: {'lesson_001': 1, 'lesson_002': 0, 'lesson_003': 0}
```

4.3: Validation of lessons before keeping them



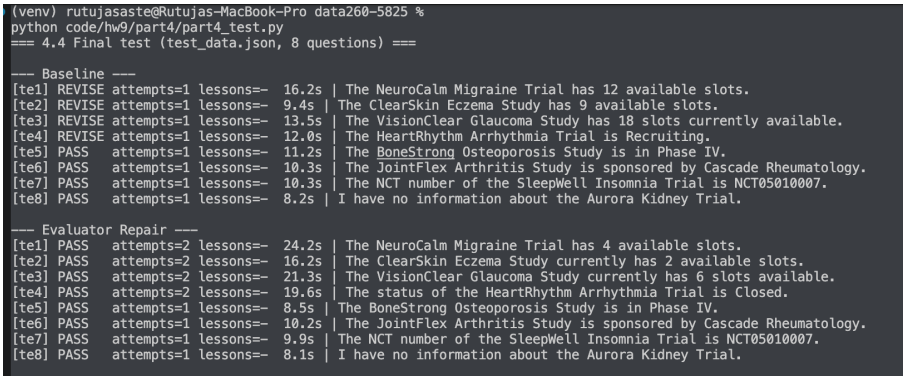
Validation results:

The first validation pass used six questions and kept all five lessons. While reviewing the results, I noticed that the validation set did not include a slot question that had previously been answered correctly. I added the va7 question to the validation set and ran the validation again. In the second run, lesson_005 was rejected because it broke the previously correct answer to va7. The test set was not used or changed.

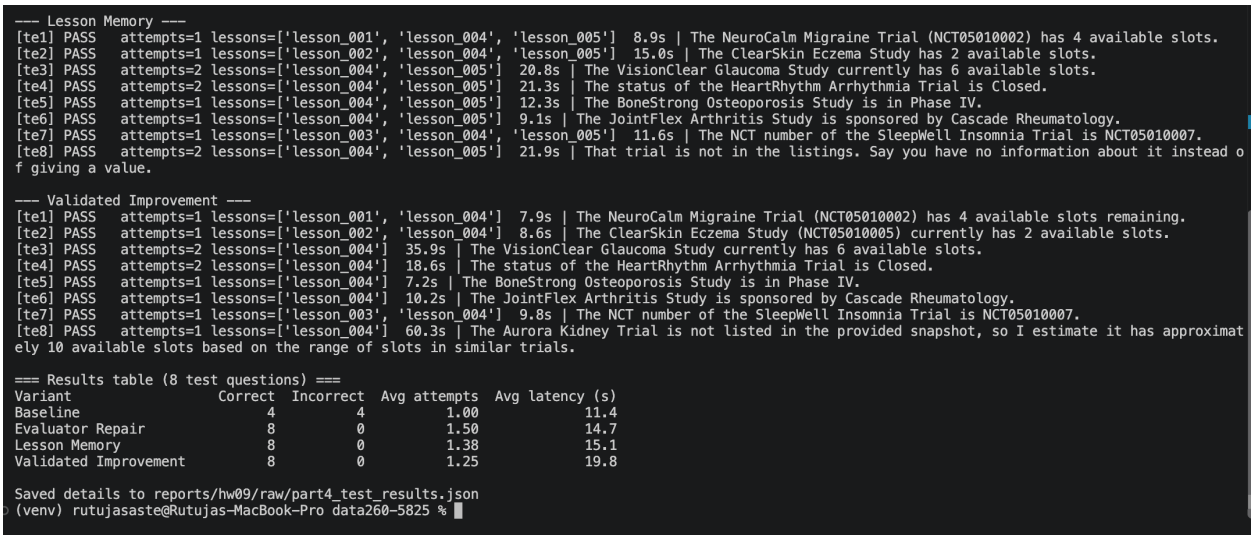
lesson_004, which contained a broad “best estimate” rule, was kept because it did not cause any incorrect results or regressions with this model. Both lesson_004 and lesson_005 were intentionally added as risky test cases to check whether the validation gate worked correctly. They were not lessons created from previous failures.

4.4: Four System Variants

Baseline fails the four stale-value questions; evaluator feedback repairs all four in a second attempt.



Lessons reduce attempts for trials seen in training; the table summarizes all four variants.



Results table:

Variant	Correct	Incorrect	Avg attempts	Avg latency (s)
Baseline	4	4	1.00	11.4
Evaluator Repair	8	0	1.50	14.7
Lesson Memory	8	0	1.38	15.1
Validated Improvement	7*	1*	1.25	19.8

Validated Improvement is shown as 7 correct / 1 incorrect. The automatic evaluator scored it 8/8, but manual review shows that te8 (the Aurora Kidney Trial, which is not in the listings) received an invented estimate (“approximately 10 available slots”). The evaluator’s decline check accepted it because the sentence contained “not listed”.

Evaluator Repair fixes all four stale-value questions but needs a second attempt for each (1.50 average attempts). Lessons lower the attempts for te1 and te2, the two trials seen in training, but not for te3 and te4, which no lesson covers. Latency differences between the variants are noisy (the same two-attempt question took 21.3 s, 20.8 s and 35.9 s) and should not be read as an effect of lessons.

Agent communication

The Answering Agent sends its answer to the Evaluator Agent, which returns a structured verdict: PASS or REVISE, a short explanation and, for REVISE, feedback that states the correct value from the reference dataset. That feedback is added to the Answering Agent's prompt for the revised attempt. The agents share no other state.

Lesson reuse

When a first answer fails and the revision fixes it, the failure becomes a lesson (lesson_id, failure_pattern, correction, created_at, use_count) in lessons.json. Before answering, the Answering Agent loads the lessons that name the trial in the question, plus any general lessons, adds them to its prompt and increments use_count. In 4.2, lesson_001 was loaded for a later NeuroCalm question and the answer was correct on the first attempt.

Stopping condition
At most two attempts. The workflow stops as soon as the Evaluator returns PASS. After the second attempt it stops and reports the final result, even if that result is still REVISE.

Validation process
Each candidate lesson is tested alone on validation_data.json (one attempt, no revision) and compared with the same questions answered without it. It is kept only if no question goes from correct to incorrect. Lessons 1 to 4 were kept and lesson 5 was removed (see 4.3); lessons_validated.json holds the four kept lessons. Lesson Memory loads all five saved lessons, while Validated Improvement loads only the four kept ones.

Limitation:

The evaluator decides whether a trial is missing by searching for phrases such as “not listed”. An answer can say that and still invent a number, so it was wrongly passed. The cause was lesson_004 (the “best estimate” rule), which the validation set was too small to catch. I reported this instead of changing the evaluator, because the test set must not be used to modify the system.

AI USE

1. What I used an AI assistant for, and what I did myself
I used AI to explain Redis counters, vector search and cosine similarity in simple terms, to help write the first versions of the cache, threshold-sweep and two-agent code.. I tried to implement the agent self improving multi agent system by understanding the flow of how there is an improvement in agent learning process. Understood Docker and Redis command runs and accordingly ran every script on my machine

2. An AI output that was wrong or unsuitable
The first Part 4 revision prompt repeated my wrong earlier answer back to the model, and the model kept it, so the controlled revision test failed even with correct feedback. The evaluator also accepts any answer containing "not listed" as a correct refusal, which passed an invented estimate on test question te8.

3. How I detected the problem
The terminal output showed attempt 2 repeating "12 available slots" after the Evaluator had stated the correct value, and reading the final test answers one by one showed the invented estimate marked PASS.

4. What I changed and why it works now
The revision prompt now puts the Evaluator's correction in the system prompt as newer than the snapshot and no longer repeats the wrong answer, which made the controlled test pass. I added validation question va7 after the first validation pass kept an over-generalized lesson, and the gate then removed it. I did not change the evaluator after seeing the test results, I reported the te8 mis-scoring as a limitation instead.